MATH 564 - Nonlinear Optimization

Today in class we experiences all manner of technical difficulties!

Here I will gather thoughts from the day ...

- Please be sure and carefully read the syllabus information and let me know if anything is unclear.
- Office hours for this class will be 10:15-11:00 MWF. (At least that is what I would like to try.)
- The students are from Math/Stats/ME/CE/ECE - a very nice mixture. Sometimes we even get students from business/econ, but not this year :(.
- The emphasis on this course is a working understanding of fundamental optimization building blocks upon which "all" algorithms are based. We will learn the language, understand the details of key algorithms, write user-friendly general-purpose code, solve sample problems provided as projects, present some results in class.
- Collaborative work on projects is ok as long as we speak on the details very soon.
- This course uses seven projects
  - Five projects that motivate the content
  - one project that is a whole-class collaboration
  - one final project as individual work ("final exam")

The optimization Problems that we will focus on can be written as follows:

minimize

objective function

$$\boxed{\begin{array}{ll} \min & f(x) \\ \text{s.t.} & x \in \Omega \subseteq \mathbb{R}^n \end{array}}$$

subject to

subset of

member of

Translated: "minimize the value of the objective function $f: \Omega \to \mathbb{R}$ over all vectors x in a feasible set specified as a subset of $\mathbb{R}^n$."

The feasible set will be expressed as

$$\boxed{\begin{array}{l} \Omega = \{x \in \mathbb{R}^n \mid g(x) \leq 0,\ h(x) = 0\} \\ g: \mathbb{R}^n \to \mathbb{R}^m, \quad h: \mathbb{R}^n \to \mathbb{R}^p \end{array}}$$

In order to answer such a question we will need to carefully understand the nature of the functions f, g, h.

Collaborative Class Exercise

READ THE DESCRIPTION ON CANVAS!

objective →

decision variables

Valve # (k):	1	2	3	4	5	6	7	8
Efficiency E_k:	16.0	32.0	11.7	23.3	14.8	24.5	8.4	53.9
Seating Angle θ_k [°]	3.19	2.11	2.86	3.07	2.95	2.20	2.95	2.62
Clearance d_k [mm]	2.50	2.10	1.61	1.56	2.02	2.13	2.01	2.50
Tension t_k [10^2N]	2.44	2.93	3.18	2.71	2.77	3.04	3.16	3.75

$$\max_{\theta,d,t} \quad E(\theta,d,t)$$
$$\text{s.t} \quad 0 \le \theta \le 5$$
$$1 \le d \le 3$$
$$1 \le t \le 4$$

The optimization problem.

(This problem can be written equivalently in the form given earlier.)

As discussed in class, together "agree" on three new test points. Provide 3 such test points every Wednesday at the start of class.

* There is a canvas page for collecting this information and for keeping some notes. *

Functions

We consider functions $f: D \to \mathbb{R}^m$ where often $D = \mathbb{R}^n$, and otherwise D is usually a simply connected closed subset of $\mathbb{R}^n$.

Def 1.1 A function f is said to be bounded above (bounded below) if $f(x) \leq M$ ($f(x) \geq m$) for all $x \in D$ and some $M \in \mathbb{R}$ ($m \in \mathbb{R}$). If f is both bounded above and bounded below, then we say that f is bounded.

examples: $f(x) = \sin(x)$ is bounded.

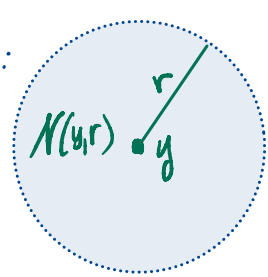
$f(x) = x^4 + x^3$ is bounded below

$f(x) = \ln x$ is bounded above on $D = (0, \infty)$

Understanding boundedness is one key to the existence of solutions. If $f(x)$ is bounded below on D, then a minimizer $y \in D$ may exist (more on this later).

Def 1.2 The neighborhood of $y \in \mathbb{R}^n$ of radius r is the set $N(y,r) = \{x \in \mathbb{R}^n \mid \|x-y\| < r\}$ for some choice of norm $\|\cdot\|$.

in $\mathbb{R}^2$:



Neighborhoods in $\mathbb{R}^n$ are important for understanding whether a point is optimal in a local sense. (more on this later)

Important Fact:

Almost all optimization algorithms are designed to find local optimal points (on restricted neighborhoods). The global problem ($r \to \infty$) is very difficult and/or very computationally expensive

Ouch! But we have some good news:

Some classes of optimization problems guarantee that locally optimal points (found by algorithms) are actually globally optimal points.

Whew! More on this later.

Def 1.3 Function f is said to be continuous at $y \in D$ if for every $\epsilon > 0$, there exists $r > 0$ such that $|f(x) - f(y)| < \epsilon$ for all $x \in N(y, r) \cap D$.

For $n=1$, f is continuous if it can be drawn without "lifting the pencil".

For $n=2$, f is continuous if surface $z = f(x)$ has no tears.

Question: Is the following function continuous at $(x,y) = (0,0)$?

$$f(x,y) = \begin{cases} 0 & \text{if } x \le 0 \text{ or } y \le 0 \\ x & \text{otherwise} \end{cases}$$

(a tear along the positive x-axis)